

Lab 31 Worksheet — Windows Networking and Repair Commands

Name: ____ Date: __

Exam objective: Use Windows networking and repair commands to diagnose connectivity and repair system file corruption (Core 2 objective 1.2).

Record as you go

Step	What you did	What you observed
1	Display the full network configuration and record the IP address, subnet mask, default gateway, DNS servers, DHCP server and MAC address.	
2	Release and renew the DHCP lease, recording the address before and after to confirm the DHCP server responded.	
3	Display and then clear the DNS resolver cache, and record when clearing it is the correct fix — after a DNS record changes but the old answer is still cached.	
4	Test the loopback address first to confirm the TCP/IP stack itself is working before testing anything external.	
5	Test the default gateway to separate a local network fault from an internet fault.	
6	Test an external IP address and then an external name, and record that success on the IP but failure on the name isolates the fault to DNS.	
7	Trace the route to a destination and identify where latency increases sharply or where the path stops.	
8	Query DNS directly and record the resolver used and the addresses returned.	

Step	What you did	What you observed
9	List active connections with the owning process ID so you can identify what is actually using the network.	
10	Run the system file checker to scan and repair protected system files, and record the result it reports.	
11	Repair the component store that sfc itself depends on, and record why DISM must be run first when sfc reports it cannot fix the files.	
12	Check the disk for file system errors, and record the difference between /f which fixes errors and /r which also locates bad sectors and takes far longer.	

Verification

Success criterion: Every command produces output you have interpreted in writing; ipconfig /all output identifies all six required fields; you correctly state that ping to IP succeeding while ping to name fails means DNS; and the sfc result is recorded with its meaning.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
